Check for updates

ACM DIGITAL LIBRARY

acm open

Latest updates: https://dl.acm.org/doi/10.1145/3627043.3659541

RESEARCH-ARTICLE

Rewriting Bias: Mitigating Media Bias in News Recommender Systems through Automated Rewriting

QIN RUAN, University College Dublin, Dublin, Leinster, Ireland

JIN XU, University College Dublin, Dublin, Leinster, Ireland

SUSAN LEAVY, University College Dublin, Dublin, Leinster, Ireland

BRIAN MAC NAMEE, University College Dublin, Dublin, Leinster, Ireland

RUIHAI DONG, University College Dublin, Dublin, Leinster, Ireland

Open Access Support provided by:

University College Dublin

PDF Download
3627043.3659541.pdf
24 February 2026
Total Citations: 3
Total Downloads: 1517

Published: 22 June 2024

Citation in BibTeX format

UMAP '24: 32nd ACM Conference on User Modeling, Adaptation and Personalization
July 1 - 4, 2024
Cagliari, Italy

Conference Sponsors:
SIGCHI
SIGWEB

UMAP '24: Proceedings of the 32nd ACM Conference on User Modeling, Adaptation and Personalization (June 2024)
https://doi.org/10.1145/3627043.3659541
ISBN: 9798400704338

Rewriting Bias: Mitigating Media Bias in News Recommender Systems through Automated Rewriting

Qin Ruan
qin.ruan@ucdconnect.ie
School of Computer Science
University College Dublin
Dublin, Ireland
Science Foundation Ireland Centre for Research Training in Machine Learning
Dublin, Ireland

Jin Xu
jin.xu1@ucd.ie
School of Computer Science
University College Dublin
Dublin, Ireland
Insight SFI Research Centre for Data Analytics
Dublin, Ireland

Susan Leavy
susan.leavy@ucd.ie
School of Information and Communication
University College Dublin
Dublin, Ireland
Insight SFI Research Centre for Data Analytics
Dublin, Ireland

Brian Mac Namee
brian.macnamee@ucd.ie
School of Computer Science
University College Dublin
Dublin, Ireland
Insight SFI Research Centre for Data Analytics
Dublin, Ireland
Science Foundation Ireland Centre for Research Training in Machine Learning
Dublin, Ireland

Ruihai Dong
ruihai.dong@ucd.ie
School of Computer Science
University College Dublin
Dublin, Ireland
Insight SFI Research Centre for Data Analytics
Dublin, Ireland

ABSTRACT

Personalised news recommender systems are effective in disseminating news content based on users' reading histories but can also amplify and proliferate biased media. This work examines the potential of automated sentence rewriting methods, utilising word replacement methods and large language models (LLMs), to mitigate this side effect of recommender systems. We present a two-step workflow: the application of automated sentence rewriting methods to rewrite biased sentences, and the integration of these rewritten sentences into the recommendation process. We evaluate the effectiveness of sentence rewriting approaches in a simulation framework, to assess how well they mitigate the spread of biased news. Our study demonstrates that applying sentence rewriting to users' reading histories can result in a significant reduction in the propagation of biased media. Our contributions are threefold: we pioneer the use of LLMs for mitigating the spread of biased news by recommender systems; we demonstrate that algorithms trained on debiased content maintain or improve recommendation accuracy; and we provide a comprehensive exploration of the effectiveness of applying sentence rewriting methods to various components within a recommender system, as well as an investigation of the underlying reasons for their efficacy. This work advances our understanding of media bias mitigation in news content and recommendation algorithms, providing valuable insights into how news recommender systems can prevent the dissemination of biased information.

This work is licensed under a Creative Commons Attribution International 4.0 License.

UMAP '24, July 01–04, 2024, Cagliari, Italy
© 2024 Copyright held by the owner/author(s).
ACM ISBN 979-8-4007-0433-8/24/07
https://doi.org/10.1145/3627043.3659541

CCS CONCEPTS

• **Information systems** → **Recommender systems**; • **Computing methodologies** → *Natural language processing*.

KEYWORDS

Media Bias, News Recommendation, Media Bias Dissemination, Debias, Sentence Rewriting

ACM Reference Format:
Qin Ruan, Jin Xu, Susan Leavy, Brian Mac Namee, and Ruihai Dong. 2024. Rewriting Bias: Mitigating Media Bias in News Recommender Systems through Automated Rewriting. In *Proceedings of the 32nd ACM Conference on User Modeling, Adaptation and Personalization (UMAP '24), July 01–04, 2024, Cagliari, Italy*. ACM, New York, NY, USA, 11 pages. https://doi.org/10.1145/3627043.3659541

1 INTRODUCTION

In the era of digital media, personalised news recommender systems have become important platforms through which people consume news [28, 39]. While these systems are widely popular for their ability to find news that aligns with user interests [28, 39], there are serious concerns about their side effects [23]. One such concern is polarisation through the propagation of biased news coverage, a

phenomenon where news content is skewed towards certain ideologies or perspectives, attacking or ignoring others [17]. Such Media bias—often arising from the perspectives of journalists, publishers and platforms—can significantly influence public opinion and discourse by presenting a distorted view of information [14]. Personalised news recommendation algorithms can propagate media bias by inadvertently creating a feedback loop, reinforcing and amplifying the pre-existing preferences of users—for example, increasing political bias and polarisation [1, 3, 12]. They also been shown to alter user beliefs and preferences to maximise user engagement [8]. Research using simulation studies demonstrates that state-of-the-art recommender systems tend to recommend more biased news for users who exhibit a preference for such content, and more neutral content for those who favour that [20, 31, 40].

In light of serious concerns about the potential negative side effects of news recommender systems [14, 23], it is vital that mechanisms to mitigate the dissemination of media bias through recommendation platforms are explored. Although the ability of news recommender systems to respond to and amplify existing biases is well documented, the potential for corrective interventions remains an open and important question. In this context, this study explores effective interventions to mitigate the proliferation of media bias in news recommender systems. In natural language processing (NLP), sentence rewriting is a widely used tool, for example in rephrasing prompts to enhance zero-shot classification model performance [21]. Our study explores the effectiveness of sentence rewriting for reducing media bias propagation in news recommender systems. Specifically, we focus on two critical research questions: (1) Are sentence rewriting methods an effective tool for reducing media bias in news content while preserving original intent and readability?, and (2) Can the use of sentence rewriting in specific components of a recommender system help reduce the propagation of media bias? To answer these research questions, we employ word replacement methods and large language models (LLMs) to automatically rewrite sentences to mitigate bias, while retaining the original context and meaning. These rewritten sentences are produced, not for presentation to users for reading, but rather for consumption by a recommendation algorithm at different stages of the recommendation process. This includes the use of sentence rewriting on users' reading histories, news articles that are in the candidate sets used in recommender systems, and news articles used to train recommendation algorithms. We use a comprehensive simulation framework to systematically assess the impact of sentence rewriting across these different aspects of the recommendation process.

Our research makes several key contributions to the field:

(1) To the best of our knowledge, we are the first to use LLMs for sentence rewriting with the clear goal of reducing media bias. We have shown that LLMs can mitigate media bias effectively and that the rewritten sentences are more readable than the original ones. We contribute a publicly accessible corpus of sentences rewritten using LLMs as a resource for the research community [34].
(2) We show that the strategic application of sentence rewriting—either to users' reading histories or to the data used to train recommendation algorithms—can mitigate bias, creating a fairer information environment for users.
(3) Our study reveals that recommendation algorithms that learn from bias-free sentences not only maintain or enhance their recommendation accuracy, but also reduce the propagation of biased news. This presents a potential method of reducing polarisation in the dissemination of news.

The remainder of the paper is organized as follows. Section 2 reviews related work on media bias and debiasing in recommendation algorithms. Section 3 introduces the sentence rewriting methods, and different scenarios for applying debiasing using sentence rewriting in a news recommender system. Section 4 describes an experiment designed to assess the proposed approaches. Section 5 analyses the quality of the rewritten sentences generated, evaluates the performance of news recommendation algorithms using these sentences, and assesses the reduction in media bias propagation achieved using sentence rewriting. Section 6 summarises the work and suggests future research directions. The source code for our methods and experiments is publicly available.[1]

2 RELATED WORK

Bias in news media is becoming increasingly apparent [14, 35] and takes various forms including selective reporting, factual cherry-picking and emotionally charged news coverage that often mirrors media sources' ideologies [2, 4, 14, 35]. Such bias is evident across the spectrum from left to right-leaning outlets and affects how stories about key societal issues are told [18]. For example, research by McKeever et al. [22] links media bias with growing anti-immigrant sentiment in regions with changing demographics, while Poberezhskaya [27] examines its role in reporting climate change. Media bias can have negative effects on society, from fostering confirmation bias and societal division to undermining trust and skewing political perceptions [5–7, 11, 26]. The role of media bias in polarizing social networks is particularly notable as it creates environments where users are exposed primarily to conforming viewpoints [1, 3, 12, 15]. News recommender systems further accelerate this trend. While recommender systems are designed to cater to individual preferences, they may also propagate media bias by choosing content that aligns with users' existing beliefs, generating an echo chamber effect and possibly limiting exposure to a wider range of perspectives [25]. Empirical research supports these observations, with simulation studies demonstrating that state-of-the-art news recommender systems tend to suggest more biased news articles to users who prefer such biased content [20, 31, 40].

Strategies to mitigate the propagation of biased media through news recommender systems are important. For example, Lee et al. [19] proposed a method that synthesises an impartial narrative by algorithmically combining the core content from multiple biased sources to create neutral summaries. Alternatively Raza et al. [29] developed Dbias, a Python package that leverages machine learning to replace biased expressions with more neutral text. It operates in a sequential manner: it first identifies biases with text, then masks biased words, and finally suggests alternative, less biased sentences. Ruan et al. [33] used the pointwise mutual information algorithm to automatically create biased and unbiased word dictionaries, and then use these to replace biased words in news headlines with appropriate neutral words. In addition to reducing bias in news

[1] https://github.com/ruanqin0706/MediaBiasinNewsRec

content, limited efforts have addressed the propagation of media bias within news recommender systems. For example, Coupette et al. [9] developed Gamine, a fast, greedy algorithm designed to minimise exposure to harmful content induced by algorithmic recommendations. This is achieved through edge rewriting in directed graphs of media items.

Research Gap Despite these advances, a significant research gap remains in mitigating media bias propagation in news recommender systems. Our study addresses this gap by debiasing news content within news recommendation algorithms using automated sentence rewriting to counter media bias propagation. By doing so, we aim to enhance the impartiality of news exposure, fostering a more informed public discourse.

3 METHODOLOGY

The rationale behind our approach to bias mitigation using sentence rewriting is that feeding debiased news articles into news recommendation algorithms is likely to weaken the likelihood of the recommender using bias queues as the basis for making recommendations. This section describes our approach, including the sentence rewriting methods used in our mitigation approach, and different strategies for using sentence rewriting in a news recommendation environment to reduce the influence of media bias in recommendation algorithms.

3.1 Sentence Rewriting Methods

Mitigating bias using sentence rewriting involves rewriting news sentences in a way that preserves their original meanings but eliminates or minimizes biased elements. We explore two approaches to sentence rewriting, one based on word replacements and a second that employs LLMs.

Sentence Rewriting Using Word Replacement We draw upon the work of Ruan et al. [33], where biased words within a sentence are identified and substituted with neutral words. Ruan et al. [33] built biased and unbiased word dictionaries, automatically generated using a pointwise mutual information (PMI) algorithm. When a biased word is found, the method looks for a neutral word with a similar meaning in the unbiased word dictionary. Replacement is guided by word similarity metrics, ensuring that the substituted words maintain the context and meaning of the original sentence. We employed five variations of de-biasing using sentence rewriting based on word replacements:

- **Replace adjectives (RADJ)**: Replaces adjectives identified as contributing to bias with neutral or opposite ones.
- **Replace adverbs (RADV)**: Replaces adverbs identified as contributing to bias with neutral or opposite ones.
- **Replace nouns (RNOUN)**: Replaces nouns identified as contributing to bias with neutral or more general ones.
- **Replace verbs (RVERB)**: Replaces verbs identified as contributing to bias with neutral or more factual ones.
- **Replace all (RALL)**: A combination of all of the above approaches.

Sentence Rewriting Using LLMs Recognising the role of language models in reshaping textual content, our methodology incorporates the use of generative pre-trained transformers, specifically GPT-3.5[2] and GPT-4.0[3] from OpenAI, for sentence rewriting tasks. The intent is to leverage these models' advanced capabilities to identify and mitigate media bias in news headlines. To this end, we designed a specific prompt aimed at guiding the models to understand the context of the sentence, recognise the presence of bias, and generate alternatives that align with the original meaning yet exhibit reduced bias indicators. This prompt below is formulated to directly prompt GPT-3.5 and GPT-4.0 to produce alternative text that remains true to the original intent and factual content while removing media bias.

The news headline: {news_headline}.
The objective is to rewrite the above news headline to eliminate any elements of media bias without altering the original meaning.
The output: {rewritten_sentence}.
Do not explain the reason or include any other words.

Based on the underlying LLM used we propose two variations of sentence rewriting using LLMs:

- **Sentence rewriting using GPT-3.5 (RG3.5)**: Rephrases sentences using GPT-3.5 based on the prompt above.
- **Sentence rewriting using GPT-4.0 (RG4.0)**: Rephrases sentences using GPT-4.0 based on the prompt above.

3.2 Integration of Rewritten News Headlines in News Recommender Systems

To study the effect of sentence rewriting on the recommendation process, seven scenarios, are explored. These scenarios are based on using sentence rewriting at different stages in the recommendation algorithm:

- applying sentence rewriting to users' reading histories when presented to the algorithm for recommendations.
- applying sentence rewriting to the news articles in the candidate news datasets presented to the recommender system to select articles to be recommended to a user.
- applying sentence rewriting to the articles in the dataset used for training news recommendation algorithms.

Each of the seven scenarios applies sentence rewriting to one or more of these stages and is designed to explore different aspects of the interaction between media bias and news recommendation algorithms. By changing the recommender system component to which sentence rewriting is applied, we can evaluate how sentence rewriting decreases the propagation of media bias and how it affects user engagement and the recommendation system's performance. The specific details for each scenario are as follows:

- **User History Integration (U)**: Sentence rewriting is applied to the users' reading histories, with all other components of the simulation remaining unchanged.
- **Candidate News Dataset Integration (C)**: Sentence rewriting is only applied to the candidate news items presented to the recommender system.
- **Dual Integration (UC)**: Sentence rewriting is applied to both user reading histories and the candidate news items presented to the recommender system.

[2]https://platform.openai.com/docs/models/gpt-3-5
[3]https://platform.openai.com/docs/models/gpt-4-and-gpt-4-turbo

- **Algorithms Trained on Rewritten News (A)**: The recommendation systems are trained using data to which sentence rewriting has been applied.
- **Algorithms Trained on Rewritten News with User History Integration (UA)**: Sentence rewriting is applied to data used to train recommender systems and to users' reading histories presented to the recommender system.
- **Algorithms Trained on Rewritten News with Candidate News Dataset Integration (CA)**: Sentence rewriting is applied to data used to train recommender systems and to the candidate news items presented to the recommender systems.
- **Algorithms Trained on Rewritten News with Dual Integration (UCA)**: Sentence rewriting is applied to the data used to train recommender systems, as well as user reading histories and candidate news items presented to the recommender system.

4 EXPERIMENTS

In this section, we describe the experimental setup designed to assess the effectiveness of the proposed sentence rewriting approaches in decreasing media bias propagation within news recommender systems. We utilised the simulation framework developed by Ruan et al. [31] as the basis for our experiments. The simulation framework can be used to generate two datasets [32]: the *News Recommendation Dataset* and the *Candidate News Dataset*. The News Recommendation Dataset includes user-news interaction records for a population of simulated users, providing a *Training Split* for training news recommendation algorithms and an *Evaluation Split* for assessing the performance of the trained news recommendation algorithms. The Candidate News Dataset includes a set of news articles, that can be presented to users while the simulation is running during experiments. The news articles in the News Recommendation Dataset and Candidate News Dataset are derived from the Hyperpartisan News Detection Dataset [17], which includes both manually labelled articles and articles labelled using distant supervision based on publisher attributes, a method that can be somewhat error-prone. To ensure accurate bias labels, we retained only those articles where the distant supervision labels matched the predictions of a state of the art bias detection model (based on the Overlap-checking (1:1) mechanism from Ruan et al. [30]). During the simulation phase of our experiments 2,000 users are simulated. These users are categorised into eight user groups based on the proportion of biased news articles in their reading history, ranging from 10% to 80%. Each group contains 250 users.

4.1 Rewriting and Evaluating Rewritten Sentences

To assess the performance of the different sentence rewriting approaches (RADJ, RADV, RVERB, RNOUN, RALL, RG3.5 and RG4.0) they were used to modify news headlines identified as biased in both the Training and Evaluation splits of the News Recommendation Dataset, as well as in the Candidate News Dataset. To ensure that the rewritten news sentences effectively mitigate media bias while preserving their original meaning and readability, we use three measures to evaluate the performance of the sentence rewriting methods:

- **De-Biased Sample Ratio (DBSR)**: The primary goal of our sentence rewriting method is to reduce media bias in news headlines. DBSR measures the proportion of sentences that are classified as unbiased after the rewriting process **U**, relative to the total number of originally biased sentences **B**.

$$DBSR = \frac{\mathbf{U}}{\mathbf{B}} \tag{1}$$

  where **U** is the number of sentences classified as unbiased after rewriting, and **B** is the total number of sentences classified as biased before rewriting. This measure indicates the method's efficacy in debiasing content.
- **Sentence Embedding Similarity (SES)**: After rewriting, sentences should keep their original meaning while becoming less biased. SES utilises BERT's CLS embeddings [16] to measure the semantic similarity of the original and rewritten sentences. Let **O** represent the BERT CLS embedding of the original sentence, and **R** represent the same for the rewritten sentence. Then, SES is given by the cosine similarity between these two embeddings. This measure assesses the semantic preservation of the rewritten text, confirming that key information and the sentences' purpose remain during the debiasing process.
- **Flesch Reading Ease (FRE)**[4]: Readability is crucial to news content, affecting its accessibility and comprehension by the general public. We utilise the Flesch Readability Score to evaluate the ease with which the rewritten sentences can be read and understood. This measure, originally developed by Flesch [13], is defined as:

$$FRE = 206.835 - 1.015\left(\frac{\text{total words}}{\text{total sentences}}\right) - 84.6\left(\frac{\text{total syllables}}{\text{total words}}\right) \tag{2}$$

  By using this measure, we ensure that the sentence rewriting process does not inadvertently make the news content more complex or difficult for readers.

4.2 Training and Evaluating News Recommendation Algorithms

We conducted experiments on three representative news recommendation algorithms: NRMS [38], PLM-Empowered [39] and FIM [37]. These news recommendation algorithms were selected due to their sensitivity to the amount of biased news articles present in users' reading histories as reported by [31]. The news recommendation algorithms (NRMS, FIM, PLM-empowered) were trained on both the original and rewritten versions of the training dataset. In the data processing phase, news titles were used as the inputs for these algorithms. The training process utilised the Adam optimizer, with a learning rate set to 0.001 and a batch size of 32. This is conducted on a server equipped with a 24GB GPU. The network was trained using the Training split and its performance is validated after specified iterations on the Validation split. Early

[4]We calculated the FRE score using the `textstat` library. See https://neon.niederlandistik.fu-berlin.de/en/textstat/ for details.

stopping based on validation loss is employed to prevent overfitting. The new recommendation algorithms were assessed on the validation split using area under the curve (*AUC*), mean reciprocal rank (*MRR*), and normalized discounted cumulative gain at ranks 5 and 10 (*NDCG@5, NDCG@10*).

4.3 Running and Evaluating Media Bias Reduction

Using the well-trained news recommender systems we conducted 65 iterations of continuous recommendations using the Candidate News Dataset. We focused on the user groups with a preference for biased news articles (70% and 80% biased articles in their initial reading history). We named these groups 'G-70%' and 'G-80%' respectively. Due to their inherent preference for biased news, these groups receive recommendations for more biased articles [31]. This is crucial as it presents potential risks of reinforcing echo chambers and intensifying information polarisation. In each iteration of the simulation, the recommendation algorithms selected 50 news articles for each user, based on their previous news interactions. We randomly picked 10 of these 50 articles and added them to the user's reading history, mimicking the user's clicks. This allowed us to observe the dynamics of news recommendation in a simulated yet realistic environment, particularly focusing on how the rewritten content influenced the media bias propagation among biased user groups. We design four measures to assist in evaluating the effectiveness of rewritten sentence methods.

- **Average Proportion of Biased Articles (APBA)** This measure is designed to assess biased news consumption for a specific user group. For each user in the group, APBA calculates the proportion of biased news articles in their reading history after one recommendation iteration. This proportion is then averaged across all users in the group. APBA is formally defined as:

$$\text{APBA}_{\text{group}} = \frac{\sum_{i=1}^{N_{\text{group}}} \frac{\text{Number of Biased News}_i}{\text{Total News}_i}}{N_{\text{group}}} \tag{3}$$

  where N_{group} represents the number of users in the group, *Number of Biased News*$_{\text{i}}$ refers to the number of biased news articles in the reading history of the i-th user after a single recommendation iteration, and *Total News*$_{\text{i}}$ is the number of news articles in a user's reading history. The $APBA_{\text{group}}$ therefore reflects the average proportion of biased news consumption per user in the given group.
- **News Theme Diversity Index (NTDI)** This measure is designed to understand the diversity of news themes in users' reading histories for a specific user group after one recommendation iteration. NTDI utilises the Shannon Diversity Index (SDI) [10] to quantify the variety of news themes in each user's reading history. The diversity index for each user is then averaged across the group to obtain the overall diversity measure. NTDI is defined as:

$$\text{NTDI}_{\text{group}} = \frac{\sum_{i=1}^{N_{\text{group}}} \text{SDI}_i}{N_{\text{group}}} \tag{4}$$

  where N_{group} represents the number of users in the group, and SDI_i is the Shannon Diversity Index calculated for the i-th user's reading history after a single recommendation iteration. The $NTDI_{\text{group}}$ thus reflects the average diversity of news themes per user in the given group, with a higher index indicating greater diversity in the range of news themes accessed.
- **Media Bias Reduction Index (MBRI)** The MBRI is designed to evaluate the effectiveness of interventions in reducing media bias propagation in news recommender systems. Firstly, the Wilcoxon signed-test [24] is used to evaluate the significant differences in the APBA curves across recommendation iterations between the original and rewritten versions for a given user group. We use a significance level of $p = 0.05$ for this assessment. Subsequently, to quantify the overall impact of the intervention, we calculate the difference in the area under the APBA curves for the original and rewritten versions using the composite trapezoidal rule. A negative value in this calculation indicates that the intervention has effectively reduced the propagation of media bias by the news recommender system.
- **Interest Diversity Impact Index (IDII)** The IDII is developed to measure the news theme diversity changes of a given user group before and after the intervention. This index applies the Wilcoxon signed-rank test [24] to compare the NTDI curves across recommendation iteration between the original and rewritten versions for a given user group. We use a significance level of $p = 0.05$ to determine whether changes in the diversity of news themes exist. Following this, the difference in the area under the NTDI curves for the original and rewritten versions is calculated to quantify the extent of diversity change. This calculation uses the composite trapezoidal rule, where a positive value represents an increase in the diversity of news themes for the given user groups.

5 RESULTS

This section describes the results of the experiments performed to assess the effectiveness of the proposed sentence rewriting approaches for bias mitigation. We first assess the effectiveness of the sentence rewriting approaches independently, focusing on their ability to remove bias without changing the meaning of sentences, then we assess the impact of sentence rewriting on the effectiveness of the recommender systems themselves to ensure that their ability to recommend relevant articles has not been damaged, before finally assessing the impact of sentence rewriting on the propagation of biased news articles within a recommender system.

5.1 The Effect of Sentence Rewriting

The three measures described in Section 4.1—De-Biased Samplee Ratio (DBSR), Sentence Embedding Similarity (SES), and Flesch Reading Ease (FRE)—are used to evaluate the performance of different sentence rewriting methods on biased news headlines with the Training and Evaluation splits of the News Recommendation Dataset and the Candidate News Dataset. The results are shown

Table 1: An evaluation of the ability of sentence rewriting approaches to remove bias (De-Biased Sample Ratio (DBSR)), while maintaining sentences' meaning (Sentence Embedding Similarity (SES)) and not reducing reading level (Flesch Reading Ease (FRE)). The arrows in the headings indicate the direction that indicates positive performance and underlines indicate best performance.

Method	Training Split			Evaluation Split			Candidate News Dataset		
	DBSR↑	SES↑	FRE↑	DBSR↑	SES↑	FRE↑	DBSR↑	SES↑	FRE↑
ORIGINAL	0.000	1.000	6.867	0.000	1.000	6.883	0.000	1.000	6.911
RADJ	2.131	0.993	6.858	2.343	0.993	6.875	2.610	0.994	6.898
RADV	1.126	0.994	6.874	1.172	0.994	6.894	1.650	0.995	6.907
RNOUN	13.470	0.971	7.119	13.339	0.971	7.112	14.464	0.976	7.225
RVERB	4.865	0.987	6.906	5.002	0.987	6.927	5.763	0.990	6.949
RALL	18.094	0.950	7.335	17.891	0.949	7.344	19.069	0.957	7.367
RG3.5	34.620	0.918	8.130	34.700	0.918	8.150	36.390	0.938	7.885
RG4.0	63.571	0.884	9.207	63.317	0.883	9.225	65.022	0.916	8.694

Table 2: The performance of news recommendation algorithms trained using original and rewritten datasets (measured using the Evaluation Split of the News Recommendation Dataset).

Algorithm	Metric	ORIGINAL	RALL	RADJ	RADV	RNOUN	RVERB	RG3.5	RG4.0
PLM-empowered	AUC↑	0.696	0.704	0.716	0.733	0.709	0.717	0.707	0.719
	MRR↑	0.266	0.269	0.276	0.277	0.275	0.281	0.278	0.275
	NDCG@5↑	0.500	0.507	0.519	0.520	0.519	0.528	0.521	0.509
	NDCG@10↑	0.557	0.562	0.576	0.581	0.574	0.580	0.572	0.567
NRSM	AUC↑	0.693	0.689	0.693	0.692	0.699	0.687	0.685	0.695
	MRR↑	0.255	0.259	0.260	0.252	0.259	0.255	0.258	0.255
	NDCG@5↑	0.476	0.480	0.487	0.468	0.478	0.476	0.482	0.475
	NDCG@10↑	0.539	0.537	0.544	0.532	0.541	0.535	0.540	0.537
FIM	AUC↑	0.605	0.631	0.633	0.615	0.635	0.626	0.619	0.634
	MRR↑	0.266	0.256	0.253	0.248	0.261	0.255	0.249	0.256
	NDCG@5↑	0.382	0.418	0.407	0.397	0.425	0.412	0.398	0.416
	NDCG@10↑	0.452	0.483	0.475	0.464	0.489	0.476	0.465	0.484

in Table 1, and it can be seen from the DBSR scores that all rewriting methods used here are effective in reducing media bias. LLM approaches (RG3.5 and RG4.0) consistently outperformed word replacement approaches in reducing media bias, as evidenced by their higher DBSR. Except for RADJ and RADV, the rewriting methods improved the readability of the sentences compared to the original news headlines. Since the rewritten sentences alter the bias detector's original decision to some degree, this implies that the bias may be related to the complexity of the sentence structures or the choice of words. The word replacement methods result in higher SES than the LLMs. This means that word replacement methods keep the original sentence structure more intact, but they are less effective in debiasing than the more comprehensive rewriting done by LLMs. In short, the results show that LLMs can not only remove media bias efficiently but also enhance the readability of the rewritten sentences. Although there is a trade-off between semantic similarity and bias reduction, the advanced capabilities of LLMs in interpreting and reformulating complex sentence structures offer a chance to achieve both objectives simultaneously.

5.2 Comparing the Performance of News Recommendations Trained on Original vs. Rewritten Datasets

The second experiment evaluated the recommender systems trained on rewritten sentences, comparing them to the recommenders trained on the original data, from the point of view of recommendation performance without regard to bias mitigation. This was to understand the impact of training on rewritten sentences on recommendation performance before considering any bias mitigation effects. Table 2 shows the performance of PLM-empowered, NRMS, and FIM measured using the Evaluation Split of the dataset used. These results indicate that the recommendation algorithms can accurately make recommendations based on rewritten sentences. In fact, in most cases recommendation performance improves after sentence rewriting. This implies that the rewriting process retains sufficient information for content understanding and processing. There are two key points revealed. Firstly, it suggests that the recommendation algorithms not only learned useful information during training but were also able to generalise well, demonstrating adaptability and robustness to different data distributions. Secondly, the high performance on the evaluation dataset implies that user experience remains unaffected, despite the algorithms being trained on

a debiased dataset. In short, this result shows that reducing bias in the training dataset does not lower the quality and effectiveness of the recommenders.

5.3 The Effect of Sentence Rewriting on Media Bias Propagation

In this section, the effects of rewriting methods on the propagation of media bias in different scenarios are analysed. First, we analyse the effect of rewriting methods on media bias propagation in different recommendation algorithms under the different scenarios described in Section 3.2. Second, projections of embedding space are used to examine the effects of the rewriting methods on the recommender systems. Finally, a comprehensive analysis of the effects of the rewriting methods on the propagation of news bias and the diversity of news themes is presented.

5.3.1 Media Bias Reduction. The first results are shown in Fig. 1. It can be observed that the sentence rewriting methods are effective when used for debiasing biased news headlines in users' reading histories (Scenario U): for all recommendation algorithms, for both groups, almost all sentence rewriting approaches lead to proportions of biased articles in articles read by users that are lower than when no sentence rewriting is used (the orange dotted line). This shows that mitigating bias in user history data can play a central role in slowing down the propagation of biased information. Furthermore, RG4.0 shows the best debiasing effect, followed by RG3.5 and RALL. However, when debiasing is applied to articles in the candidate news datasets only (Scenario C), the sentence rewriting approaches were not as effective. Debiasing biased news headlines in the candidate news items resulted in similar trends to when no sentence rewriting is used. One possible reason is that the rewritten and original biased news headlines are still too close in the embedding space to influence the decision-making of the recommendation algorithms.

When sentence rewriting was used for debiasing on both user reading histories and candidate news items simultaneously (scenario UC), the effect was similar to debiasing user reading histories alone. This suggests that debiasing user reading history data plays a major role in recommending fewer biased news articles. When recommender systems were trained using the debiased data (Scenario A), a general reduction in the amount of media bias in recommended articles was observed, confirming that training using rewritten sentences is an effective debiasing strategy.

However, when training using rewritten sentences was combined with debiasing user reading histories (Scenario UA), the results were mixed. For PLM-empowered and FIM, there is some effective reduction and some increases in media bias propagation. For NRMS, the bias consistently increased. These results reveal complex interactions between the internal structure of the algorithms and the debiasing processes. On the other hand, the effect of debiasing the candidate news items along with the data used to train the recommender system (Scenario CA) was similar to the effect when only training on debiased articles (Scenario A) was used. This further demonstrates that the process of debiasing candidate news items has limited effects on the amount of biased news recommended to users.

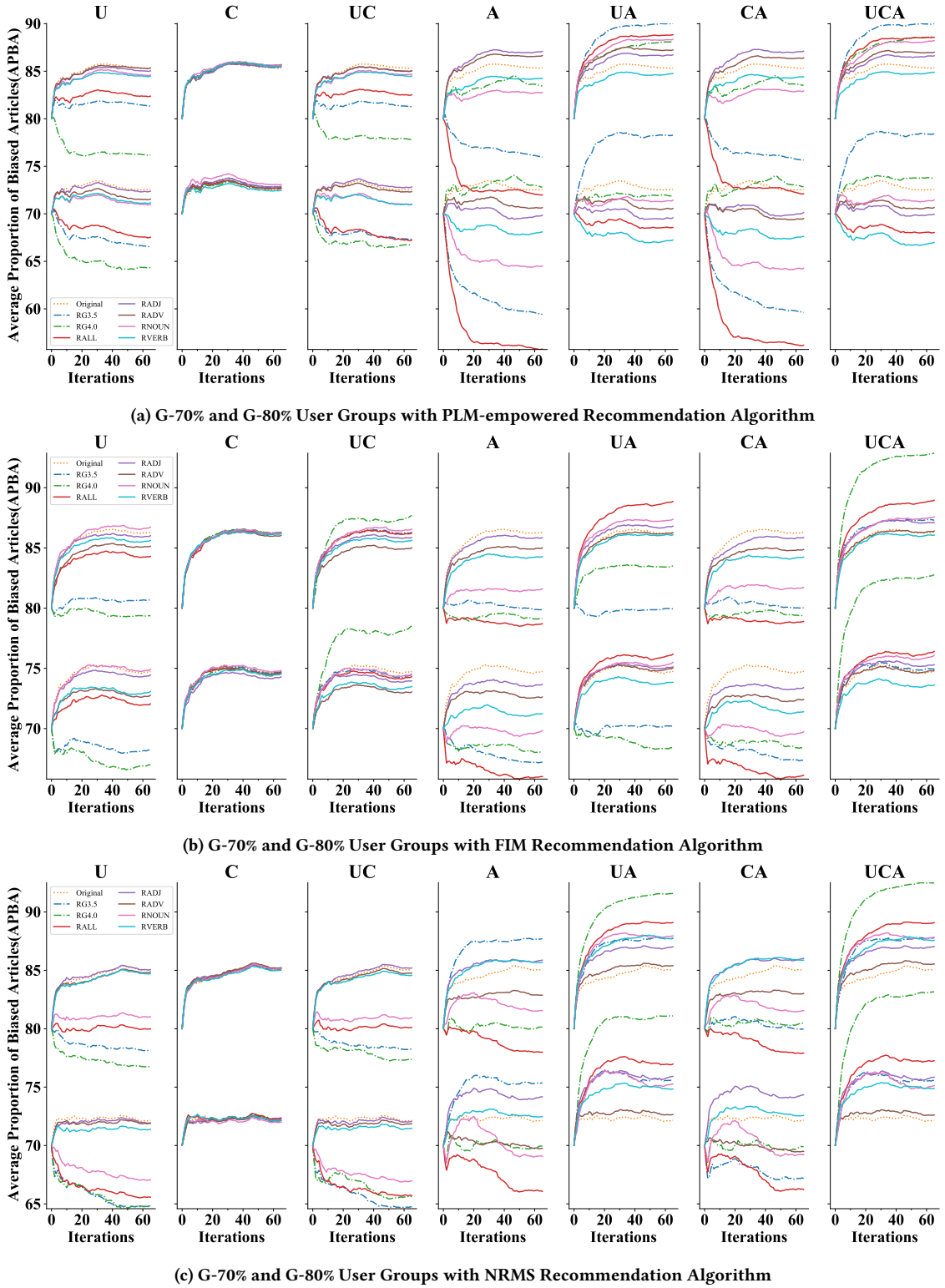
Ultimately, when debiasing was applied across all possible ways (Scenario UCA), PLM-empowered and FIM showed some reduction in media bias, but also instances where the proportion of biased news in recommended items increased; for NRMS, the proportion of biased news in recommended items consistently rose. These results are consistent with the strategy of simultaneously training with debiased data and debiasing user reading histories. This phenomenon may point to an overfitting effect where recommendation algorithms may exhibit unstable responses to multiple debiasing operations that warrant further exploration.

In summary, analysis reveals that debiasing articles using rewriting only for candidate news items (Scenario C) consistently shows no impact on bias propagation, while debiasing using sentence rewriting for articles in users' reading histories (Scenarios U and UC) consistently helps reduce the proportion of biased articles in items recommended to users. On the other hand, when news articles debiased using sentence rewriting are used to train recommender systems without changing users' reading histories (Scenarios A and CA), these systems demonstrate substantial capability in mitigating bias propagation in the news recommender system. However, when debiasing is also applied to users' reading histories (Scenarios AU and UCA) this tends to significantly exacerbate the bias issue, highlighting the need for careful consideration of strategies selection in bias mitigation strategies.

5.3.2 Embedding Space Exploration. This section uses explorations of embedding space to investigate the effects of sentence rewriting methods applied at different phases in the recommendation process. Both NRMS and PLM-empowered operate on a two-tower model, where a news encoder translates news headlines into news embeddings, and a user encoder processes the embeddings of the user's reading history into a user embedding. These embeddings are compared using a dot product similarity score to assess the likelihood of a user clicking on a candidate news article. FIM takes a more complex approach, involving similarity calculations between the user embedding and all candidate news embeddings, followed by a trainable prediction layer to output probabilities. However, fundamentally, it still computes the similarity between candidate news and the user.

Effective sentence rewriting methods imply that the similarity calculation made by the recommendation algorithms between user embeddings and unbiased candidate news embeddings will increase (reduced distance), and decrease between user embeddings and biased candidate news embeddings (increased distance). Therefore, we project the generated user and news embeddings during the prediction process onto a 2D space to observe changes in distances, providing a visual representation of the debiasing effect on the recommendation algorithm's decision-making. Fig. 2 shows this using the RG4.0 rewriting method and the user and news embeddings provided by PLM-empowered for the G-80% user group. Embeddings were projected into a 2D space using t-SNE [36] for visualization.

Fig. 2a compares the changes between the original dataset and the dataset where biased news headlines in the user reading history are debiased. In the original dataset, the average distance between the user centroid and the biased candidate news is 54.84, closer than to the unbiased news at 61.55, indicating a higher likelihood of users clicking on biased news articles. After debiasing the user

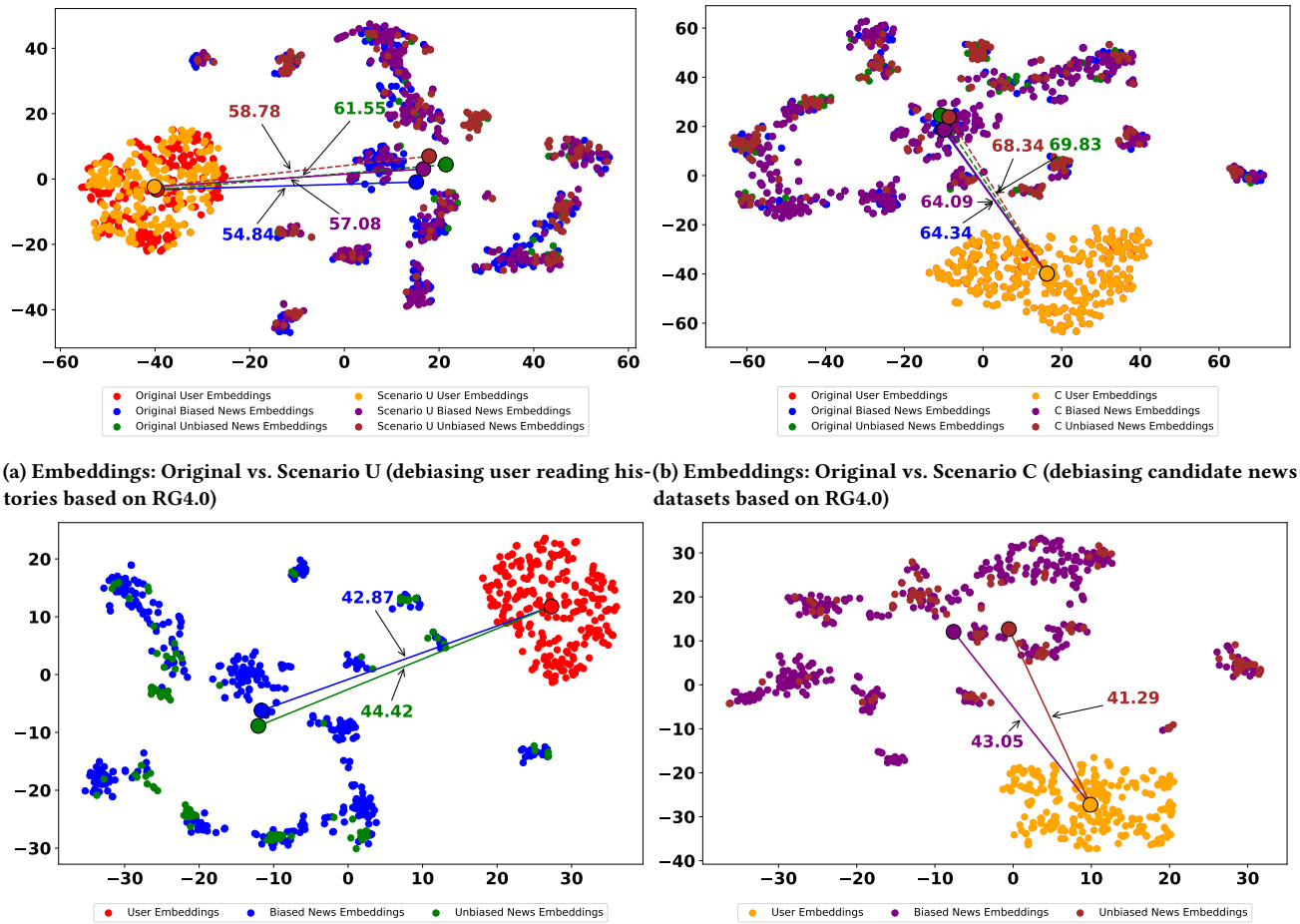


(a) G-70% and G-80% User Groups with PLM-empowered Recommendation Algorithm

(b) G-70% and G-80% User Groups with FIM Recommendation Algorithm

(c) G-70% and G-80% User Groups with NRMS Recommendation Algorithm

Figure 1: (a)-(c) show the relationship between recommendation iterations (horizontal axis) and the APBA (vertical axis) for PLM-empowered, FIM, NRMS across seven debiasing scenarios (U, C, UC,A, UA, CA, UCA) using sentence rewriting methods (RADJ, RADV, RVERB, RNOUN, RALL, RG3.5, RG4.0). The orange dotted line represents the control without any intervention applied.



(a) Embeddings: Original vs. Scenario U (debiasing user reading histories based on RG4.0)

(b) Embeddings: Original vs. Scenario C (debiasing candidate news datasets based on RG4.0)

(c) Embeddings generated by the PLM-empowered trained on the original dataset

(d) Embeddings generated by the PLM-empowered trained on the RG4.0 rewritten dataset

Figure 2: Impact of rewriting methods on news recommendation embeddings

reading histories, the distances to the biased and unbiased news became 57.08 and 58.78, increasing the likelihood of users clicking on unbiased news.

Fig. 2b compares the changes between the original dataset and the dataset with debiased biased candidate news. Initially, the distances between the user and biased/unbiased news are 64.34 and 69.83. With debiased candidate news, these distances change to 64.09 and 68.34. This small change further suggests that altering the news content itself does not significantly impact the embedding to cause a change in recommendations.

Figs. 2c and 2d demonstrate the changes in embeddings when the recommender system was trained using original texts, Fig. 2c, and text debiased using RG4.0, Fig. 2d. Before using RG4.0, distances from the user embedding to unbiased and biased news are 44.42 and 42.87. After training with the text rewritten using RG4.0, these distances shift to 41.29 and 43.05, demonstrating the effectiveness of training with sentences rewritten using RG4.0 in mitigating the algorithm's learning of media bias patterns for PLM-empowered.

5.3.3 News Theme Diversity. The last experiment looks beyond the reduction of media bias, using the Shannon diversity index to measure the diversity of themes in a user's reading history. A higher diversity of themes implies that users engage with a broader spectrum of news themes. The experiment analysed 294 test conditions across three news recommendation algorithms, two user groups, seven scenarios, and seven rewriting methods, applying measures MBRI and IDII (see Section 4.3) to determine if the conditions significantly altered media bias proportion and diversity and whether the change was positive or negative as shown in Table 3.

Out of 294 test results, in 181 instances (61.6%), recommendation bias was reduced, in 105 instances (35.7%), it was increased, and in 8 instances (2.7%) there was no significant impact. This also demonstrates the overall efficacy of the applied rewriting methods in addressing bias within the recommendation algorithms. Among the instances that reduced recommendation bias, applying debiasing strategies also improved the diversity of news themes in 101 instances (55.8%), had no effect on theme diversity in 10 instances (5.5%), and decreased theme diversity in 70 instances (38.7%). This

Table 3: Media bias proportion and news theme diversity

ALG.	Group	Method	U		C		UC		A		UA		CA		UCA	
			MBRI	IDII	MBRI	IDII	MBRI	IDII	MBRI	IDII	MBRI	IDII	MBRI	IDII	MBRI	IDII
PLM.	G-70%	RADJ	-10.259	0.697	22.486	0.636	15.861	1.574	-167.114	-2.641	-176.014	-0.231	-153.431	-1.129	-156.398	0.257
		RADV	-49.487	0.802	8.780	0.353	-7.404	1.161	-105.914	2.757	-111.856	1.917	-173.953	2.190	-120.019	1.943
		RALL	-279.933	3.330	10.363	0.330	-300.495	3.376	-990.501	-3.065	-245.842	0.612	-965.881	-2.585	-279.340	0.611
		RG3.5	-344.635	-0.803	19.741	0.488	-312.649	-1.792	-745.917	1.832	302.201	-1.404	-737.796	1.623	313.454	-0.553
		RG4.0	-497.737	-0.521	4.493	0.734	-375.791	2.090	14.267	-1.436	-59.063	5.228	9.699	-1.259	48.511	4.599
		RVERB	-85.079	2.319	43.185	0.083	-76.192	2.314	-480.031	4.639	-93.700	6.366	-494.794	5.284	-85.781	5.711
		RNOUN	-76.448	-0.340	-8.600	-0.139	-81.857	0.368	-282.791	1.636	-329.610	1.268	-309.870	1.581	-345.941	0.758
	G-80%	RADJ	-16.674	-0.493	-0.379	-0.784	-14.481	-0.900	85.093	-2.199	65.868	-0.719	87.503	-1.961	53.730	-1.494
		RADV	-8.000	0.039	10.411	-0.949	-14.516	-0.361	61.480	1.877	102.705	1.216	46.808	1.288	80.384	1.913
		RALL	-169.640	2.542	11.573	-0.882	-156.668	2.071	-780.257	-4.430	178.668	2.523	-764.852	-5.324	164.935	2.279
		RG3.5	-227.484	-3.142	9.388	-0.503	-233.101	-2.321	-523.742	-1.967	241.717	-2.891	-554.544	-1.573	242.951	-2.954
		RG4.0	-547.942	-3.379	0.879	-0.910	-450.183	-1.148	-106.244	-10.595	120.315	5.818	-107.892	-10.440	150.400	4.134
		RVERB	-40.570	1.313	17.418	-0.252	-40.382	1.591	-162.982	0.892	148.458	2.544	-161.228	1.042	133.387	2.377
		RNOUN	-51.610	-0.569	10.755	-0.552	-45.051	-0.610	-81.518	0.167	-53.525	0.010	-66.819	0.472	-48.962	0.361
FIM	G-70%	RADJ	-23.032	0.317	-33.638	-0.067	-39.219	0.876	-69.093	-2.603	12.628	-1.986	-84.750	-2.307	27.247	-1.863
		RADV	-116.083	0.250	-6.644	-0.173	-99.757	0.732	-119.149	-1.552	5.918	-2.497	-140.792	-1.573	-1.800	-2.715
		RALL	-156.608	3.436	-10.380	-0.181	-27.785	-0.213	-520.882	-0.943	56.175	-2.659	-517.281	-1.695	68.675	-2.382
		RG3.5	-397.595	0.510	-1.742	-0.374	-14.688	-0.579	-432.155	1.710	-291.459	5.511	-423.164	1.678	12.450	-3.531
		RG4.0	-467.329	1.954	-14.320	-0.013	173.983	1.115	-395.626	5.012	-363.776	0.302	-372.269	5.001	433.175	-3.119
		RVERB	1.556	0.647	4.529	-0.107	-13.319	0.811	-311.141	-1.290	20.981	-1.502	-308.017	-1.435	54.072	-2.263
		RNOUN	-105.014	0.770	-18.028	-0.216	-82.563	-0.038	-211.771	-1.750	-58.265	-2.292	-188.575	-2.035	-71.445	-2.405
	G-80%	RADJ	-15.654	-0.079	-4.386	-0.294	-25.939	0.102	-26.667	-2.107	28.042	-2.543	-35.251	-1.720	49.595	-2.308
		RADV	-72.509	1.632	-8.094	0.128	-78.285	1.267	-79.847	-1.473	-8.293	-1.516	-84.830	-1.318	-0.021	-2.142
		RALL	-108.078	3.576	-5.156	-0.102	-4.182	-0.111	-449.487	0.885	119.991	-2.698	-442.020	0.593	128.619	-2.382
		RG3.5	-337.732	1.784	2.708	0.049	-0.031	0.153	-359.358	3.151	-389.349	6.245	-351.180	3.350	55.414	-2.720
		RG4.0	-403.954	3.162	4.390	0.229	55.496	2.163	-419.997	6.607	-166.355	0.523	-404.492	6.020	357.352	-1.966
		RVERB	17.297	0.685	0.289	0.821	7.871	0.653	-282.585	0.779	53.210	-1.315	-270.717	0.599	55.401	-1.803
		RNOUN	-37.577	0.788	-4.164	0.423	-39.958	0.939	-118.925	-1.353	-19.463	-1.509	-122.591	-1.244	-17.604	-1.901
NRMS	G-70%	RADJ	-16.262	1.697	-8.243	0.032	-13.694	1.574	121.850	-4.873	218.388	-3.534	132.327	-5.293	213.311	-4.264
		RADV	-22.838	-0.081	-8.395	-0.182	-28.550	-0.094	-131.779	-0.474	30.645	-0.475	-150.110	-0.004	28.805	-0.680
		RALL	-383.762	9.527	5.087	-0.319	-380.822	9.662	-306.692	-1.402	277.794	4.386	-298.884	-0.973	286.810	4.039
		RG3.5	-421.248	-0.691	-2.540	-0.059	-421.027	-0.527	182.666	-1.928	209.119	-2.215	-286.850	-0.377	200.620	-2.177
		RG4.0	-422.636	0.067	-2.233	-0.048	-371.871	0.058	-148.489	9.364	499.714	-5.652	-152.867	9.892	601.426	-4.690
		RVERB	-289.496	6.916	-13.189	-0.101	-303.072	7.072	-119.417	-6.837	195.981	-1.505	-124.581	-6.849	188.633	-1.532
		RNOUN	-57.487	1.766	1.986	-0.442	-49.449	1.249	20.256	-3.695	155.309	-3.721	33.390	-4.354	160.276	-3.374
	G-80%	RADJ	6.323	1.685	12.052	0.246	9.364	2.071	54.342	-3.260	121.073	-1.782	56.486	-2.864	119.341	-1.664
		RADV	-15.129	0.503	14.912	-0.121	-10.345	0.826	-109.929	0.904	29.417	0.602	-101.492	0.889	38.739	0.256
		RALL	-290.741	13.054	7.180	0.316	-286.442	13.046	-362.614	2.772	229.436	6.135	-368.890	2.678	225.381	6.423
		RG3.5	-385.777	1.697	4.402	-0.319	-381.930	1.829	158.457	0.424	157.859	0.329	-267.878	2.339	159.857	0.185
		RG4.0	-467.958	1.200	7.413	0.159	-429.248	1.239	-281.968	9.966	361.715	-2.456	-264.028	9.805	413.395	-3.225
		RVERB	-228.853	10.258	4.788	-0.025	-236.634	9.925	-157.081	-5.266	178.186	-1.320	-165.692	-5.119	176.665	-0.850
		RNOUN	-19.377	2.029	-0.148	0.057	-20.127	2.018	46.348	-3.465	163.803	-3.693	58.421	-3.674	152.929	-3.868

(1) Black text indicates reduced media bias or enhanced diversity, blue text signifies no change, and orange text denotes increased media bias or reduced diversity.
(2) PLM. stands for PLM-empowered. ALG. stands for Algorithm.

suggests that users are being exposed to a broader spectrum of news topics, enhancing the informativeness and breadth of news dissemination. Specifically, in the scenario where users' reading history was debiased (Scenario U), 76.9% of the methods managed to lower bias and maintain or increase theme diversity. This indicates that the strategic application of debiasing techniques can lead to more balanced recommendations without sacrificing content diversity.

In short, our experimental results indicate that rewriting methods can mitigate bias without limiting the diversity of recommender systems. In some scenarios (e.g., Scenario U, Scenario UC) rewriting methods can also improve the diversity of recommender systems, which may be the result of the sentence rewriting methods that enable recommender systems to obtain more semantic information.

6 CONCLUSION

In this work, automated sentence rewriting methods using word replacement (RADJ, RADV, RALL, RVERB, RNOUN) and LLMs (RG3.5, RG4.0) were used to mitigate the propagation of media bias in news recommender systems. To the best of our knowledge, this is the first time that LLMs have been used for sentence rewriting to reduce media bias. In addition, these rewriting methods were implemented across seven distinct application scenarios within a simulated news consumption environment. Sentence rewriting can be applied to users' reading history, to the candidate news dataset used during simulation, or to the data used to train the recommender systems themselves—as well as combinations of these. This research contributes to the ongoing discourse on media bias in news recommender systems by highlighting the feasibility of using

sentence rewriting methods (both LLM and word replacement methods) to debias news content. The results show that the propagation of media bias can be reduced by debiasing news articles in users' reading history and datasets for training news recommendation algorithms. This approach of using debiasing to alleviate the extent to which news recommendation algorithms catalyse the spread of media bias not only opens new avenues for the fair dissemination of news but also provides debiased and neutral data resources for future research. In the future, we plan to refine the design of recommendation algorithms to enhance the objectivity and fairness of news consumption.

ACKNOWLEDGMENTS

This publication has emanated from research conducted with the financial support of the EU Commission Recovery and Resilience Facility under the Science Foundation Ireland OurTech Challenge Grant Number 22/NCF/OT/11077, and Science Foundation Ireland under Grant numbers 18/CRT/6183 and 12/RC/2289_P2. For the purpose of Open Access, the author has applied a CC BY public copyright licence to any Author Accepted Manuscript version arising from this submission.

REFERENCES

[1] Eytan Bakshy, Solomon Messing, and Lada A Adamic. 2015. Exposure to ideologically diverse news and opinion on Facebook. *Science* 348, 6239 (2015), 1130–1132.
[2] Ramy Baly, Giovanni Da San Martino, James Glass, and Preslav Nakov. 2020. We Can Detect Your Bias: Predicting the Political Ideology of News Articles. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*. 4982–4991.
[3] Pablo Barberá, John T Jost, Jonathan Nagler, Joshua A Tucker, and Richard Bonneau. 2015. Tweeting from left to right: Is online political communication more than an echo chamber? *Psychological science* 26, 10 (2015), 1531–1542.
[4] Tony Bennett. 2005. Media,'reality', signification. In *Culture, society and the media*. Routledge, 285–306.
[5] Dan Bernhardt, Stefan Krasa, and Mattias Polborn. 2008. Political polarization and the electoral effects of media bias. *Journal of Public Economics* 92, 5-6 (2008), 1092–1104.
[6] Maxwell T Boykoff. 2007. Flogging a dead norm? Newspaper coverage of anthropogenic climate change in the United States and United Kingdom from 2003 to 2006. *Area* 39, 4 (2007), 470–481.
[7] Heinz Brandenburg. 2006. Party strategy and media bias: A quantitative analysis of the 2005 UK election campaign. *Journal of elections, public opinion and parties* 16, 2 (2006), 157–178.
[8] Micah Carroll, Dylan Hadfield-Menell, Stuart Russell, and Anca Dragan. 2021. Estimating and penalizing preference shift in recommender systems. In *Proceedings of the 15th ACM Conference on Recommender Systems*. 661–667.
[9] Corinna Coupette, Stefan Neumann, and Aristides Gionis. 2023. Reducing exposure to harmful content via graph rewiring. In *Proceedings of the 29th ACM SIGKDD Conference on Knowledge Discovery and Data Mining*. 323–334.
[10] Tommaso Di Noia, Vito Claudio Ostuni, Jessica Rosati, Paolo Tomeo, and Eugenio Di Sciascio. 2014. An analysis of users' propensity toward diversity in recommendations. In *Proceedings of the 8th ACM Conference on Recommender Systems*. 285–288.
[11] Caroline Fisher, Terry Flew, Sora Park, Jee Young Lee, and Uwe Dulleck. 2021. Improving trust in news: Audience solutions. *Journalism Practice* 15, 10 (2021), 1497–1515.
[12] Seth Flaxman, Sharad Goel, and Justin M Rao. 2016. Filter bubbles, echo chambers, and online news consumption. *Public opinion quarterly* 80, S1 (2016), 298–320.
[13] Rudolph Flesch. 1948. A new readability yardstick. *Journal of applied psychology* 32, 3 (1948), 221.
[14] Felix Hamborg, Karsten Donnay, and Bela Gipp. 2019. Automated identification of media bias in news articles: an interdisciplinary literature review. *International Journal on Digital Libraries* 20, 4 (2019), 391–415.
[15] Ruben Interian, Ruslán G. Marzo, Isela Mendoza, and Celso C Ribeiro. 2022. Network polarization, filter bubbles, and echo chambers: an annotated review of measures and reduction methods. *International Transactions in Operational Research* (2022).
[16] Jacob Devlin Ming-Wei Chang Kenton and Lee Kristina Toutanova. 2019. Bert: Pre-training of deep bidirectional transformers for language understanding. In *Proceedings of naacL-HLT*, Vol. 1. 2.
[17] Johannes Kiesel, Maria Mestre, Rishabh Shukla, Emmanuel Vincent, Payam Adineh, David Corney, Benno Stein, and Martin Potthast. 2019. Semeval-2019 task 4: Hyperpartisan news detection. In *Proceedings of the 13th International Workshop on Semantic Evaluation*. 829–839.
[18] Vivek Kulkarni, Junting Ye, Steven Skiena, and William Yang Wang. 2018. Multi-view Models for Political Ideology Detection of News Articles. In *Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing*. 3518–3527.
[19] Nayeon Lee, Yejin Bang, Tiezheng Yu, Andrea Madotto, and Pascale Fung. 2022. NeuS: Neutral Multi-News Summarization for Mitigating Framing Bias. In *Proceedings of the 2022 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*. 3131–3148.
[20] Ping Liu, Karthik Shivaram, Aron Culotta, Matthew A Shapiro, and Mustafa Bilgic. 2021. The interaction between political typology and filter bubbles in news recommendation algorithms. In *Proceedings of the Web Conference 2021*. 3791–3801.
[21] Jinghui Lu, Dongsheng Zhu, Weidong Han, Rui Zhao, Brian Mac Namee, and Fei Tan. 2023. What Makes Pre-trained Language Models Better Zero-shot Learners?. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*. 2288–2303.
[22] Brooke Weberling McKeever, Daniel Riffe, and Francesca Dillman Carpentier. 2012. Perceived hostile media bias, presumed media influence, and opinions about immigrants and immigration. *Southern Communication Journal* 77, 5 (2012), 420–437.
[23] Eliza Mitova, Sina Blassnig, Edina Strikovic, Aleksandra Urman, Aniko Hannak, Claes H de Vreese, and Frank Esser. 2023. News recommender systems: A programmatic research review. *Annals of the International Communication Association* 47, 1 (2023), 84–113.
[24] Peter C O'Brien and Thomas R Fleming. 1987. A paired Prentice-Wilcoxon test for censored paired data. *Biometrics* (1987), 169–180.
[25] Eli Pariser. 2011. *The filter bubble: What the Internet is hiding from you*. Penguin, London.
[26] Uwe Peters. 2022. What is the function of confirmation bias? *Erkenntnis* 87, 3 (2022), 1351–1376.
[27] Marianna Poberezhskaya. 2015. Media coverage of climate change in Russia: Governmental bias and climate silence. *Public Understanding of Science* 24, 1 (2015), 96–111.
[28] Shaina Raza and Chen Ding. 2022. News recommender system: a review of recent progress, challenges, and opportunities. *Artificial Intelligence Review* (2022), 1–52.
[29] Shaina Raza, Deepak John Reji, Dora D Liu, Syed Raza Bashir, and Usman Naseem. 2022. An Approach to Ensure Fairness in News Articles. *arXiv preprint arXiv:2207.03938* (2022).
[30] Qin Ruan, Brian Mac Namee, and Ruihai Dong. 2021. Bias Bubbles: Using Semi-Supervised Learning to Measure How Many Biased News Articles Are Around Us.. In *AICS*. 153–164.
[31] Qin Ruan, Brian Mac Namee, and Ruihai Dong. 2023. Unveiling the Relationship Between News Recommendation Algorithms and Media Bias: A Simulation-Based Analysis of the Evolution of Bias Prevalence. In *International Conference on Innovative Techniques and Applications of Artificial Intelligence*. Springer, 210–215.
[32] Qin Ruan, Brian Mac Namee, and Ruihai Dong. 2024. *Media Bias Aware Simulation Dataset*. https://doi.org/10.5281/zenodo.11168981
[33] Qin Ruan, Brian Mac Namee, and Ruihai Dong. 2023. Reducing Media Bias in News Headlines. In *2023 31st Irish Conference on Artificial Intelligence and Cognitive Science (AICS)*. 1–4. https://doi.org/10.1109/AICS60730.2023.10470850
[34] Qin Ruan, Jin Xu, Susan Leavy, Brian Mac Namee, and Ruihai Dong. 2024. *Rewritten Media Bias News Headlines*. https://doi.org/10.5281/zenodo.11162894
[35] Adam J Schiffer. 2017. Evaluating media bias. (2017).
[36] Laurens Van der Maaten and Geoffrey Hinton. 2008. Visualizing data using t-SNE. *Journal of machine learning research* 9, 11 (2008).
[37] Heyuan Wang, Fangzhao Wu, Zheng Liu, and Xing Xie. 2020. Fine-grained interest matching for neural news recommendation. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*. 836–845.
[38] Chuhan Wu, Fangzhao Wu, Suyu Ge, Tao Qi, Yongfeng Huang, and Xing Xie. 2019. Neural news recommendation with multi-head self-attention. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*. 6389–6394.
[39] Chuhan Wu, Fangzhao Wu, Yongfeng Huang, and Xing Xie. 2023. Personalized news recommendation: Methods and challenges. *ACM Transactions on Information Systems* 41, 1 (2023), 1–50.
[40] Han Zhang, Ziwei Zhu, and James Caverlee. 2023. Evolution of Filter Bubbles and Polarization in News Recommendation. In *Advances in Information Retrieval: 45th European Conference on Information Retrieval, ECIR 2023, Dublin, Ireland, April 2–6, 2023, Proceedings, Part II*. Springer, 685–693.

Received 29 January 2024; accepted 25 March 2009; revised 09 May 2009